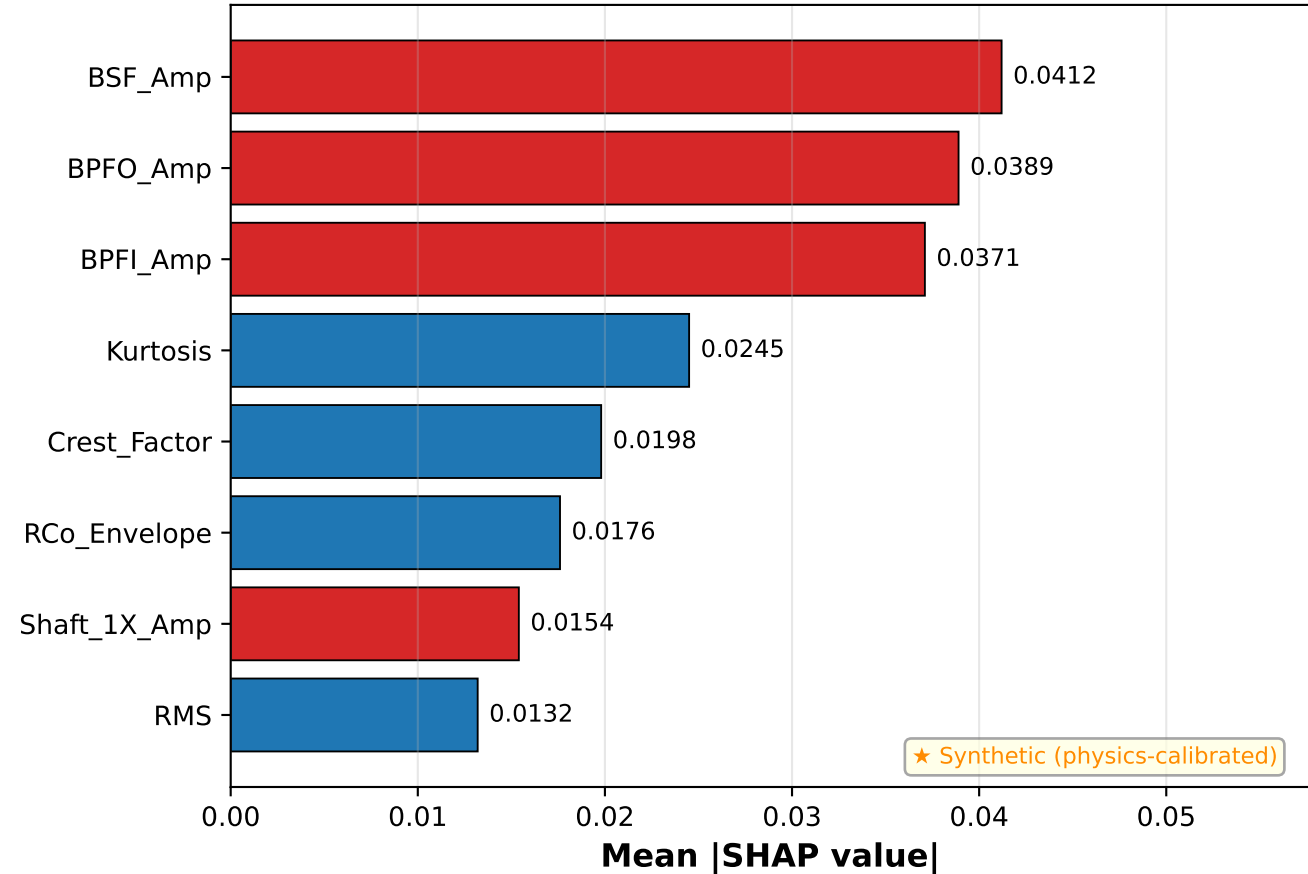
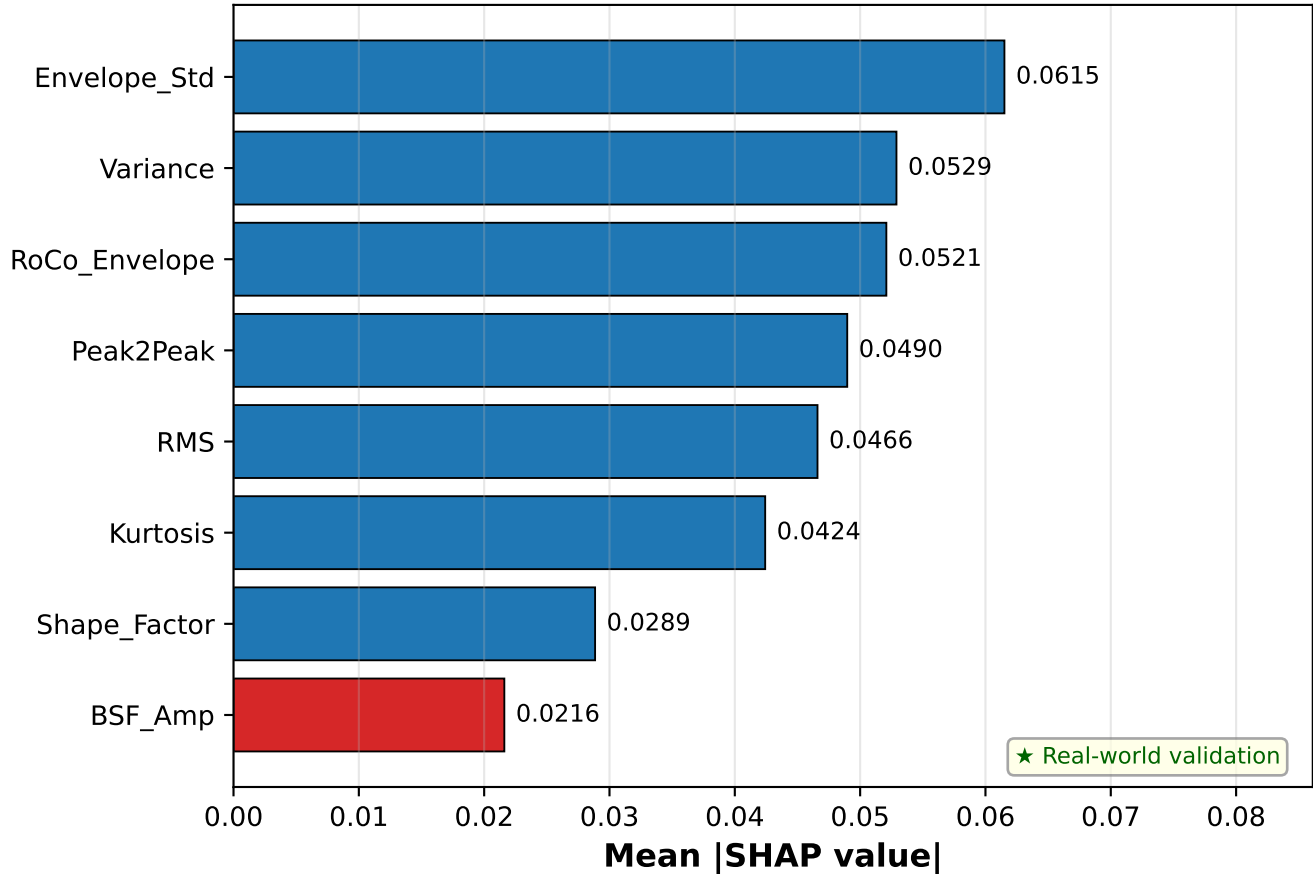


Physics-Alignment Criterion: Synthetic → Real Transfer
Fault-characteristic spectral features (red) dominate in both domains,
confirming the criterion holds on real-world data without pipeline modification.

Synthetic Bearing Data
(Parametric simulator, BPFI/BPFO features dominant)



Real CWRU Bearing Data
(Field vibration recordings, same pipeline, no tuning)



■ Fault-characteristic spectral (BPFI/BPFO/BSF) ■ Time-domain / other